Predicting High Electricity Use in Sri Lankan Households with an Explainable Boosting Machine

Applied Machine Learning: Assignment Report

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Code: https://github.com/niwanthaik/household-electricity-ebm

Live app: https://household-electricity.redwave-d9ae0d05.eastasia.azurecontainerapps.io

Summary. Using a real survey of 3,983 Sri Lankan households linked to their electricity-provider records (ReCon SL), this project predicts, from survey answers alone, whether a household's typical monthly consumption exceeds 90 kWh, the domestic tariff boundary above which the unit price and fixed charge rise sharply. An **Explainable Boosting Machine (EBM)** reaches a test ROC-AUC of **0.822** (95% CI 0.792–0.849). That is statistically indistinguishable from random forest, XGBoost, LightGBM and logistic regression, and clearly better than a decision tree, while being fully interpretable and overfitting far less than the tree ensembles. Its explanations are cross-checked with SHAP and LIME.

1. Problem definition and dataset

1.1 Problem

Sri Lanka's domestic electricity tariff is block-based. Under the PUCSL domestic tariff, consumption above 90 units a month moves a household into blocks with a much higher unit charge and fixed charge [6]. Utilities (CEB, LECO) and energy-efficiency or solar programmes want to know *which* households are likely to exceed that line, and *why*, so advice and support can be targeted. The task is **binary classification**: *will a household's median monthly use exceed 90 kWh?*, predicted only from information collected in a household survey, before any bill is seen.

1.2 Dataset

ReCon SL: Residential Electricity Consumption Dataset, Sri Lanka (IEEE DataPort, open access) [1]. It combines utility billing records for 4,063 households (101,575 monthly readings, Oct 2022 – Oct 2024) with three waves of household surveys. This project uses the Wave 1 survey (conducted 2023-12-09 to 2024-10-24): household information, demographics, and appliance, room, AC, fan and light rosters. The households are mainly in Western and Southern Province service areas (Moratuwa, Panadura, Galle, Kalutara, Maharagama and others).

1.3 Target, features and the timing rule

Target: the median of non-zero monthly readings in the 12 months up to the survey month; label 1 if above 90 kWh. 80 households with fewer than six valid months or mostly zero readings (likely vacant) were removed, leaving **3,983** households; **54%** are high users (median 96 kWh; quartiles 63 / 146), so the classes are balanced.

Features (38): the multi-row rosters were aggregated to one row per household: household size, children, elderly members, members working, hours at home; floor area, rooms, bedrooms, building age, house type; number and weekly hours of ACs, fans and lights (lighting kWh = wattage × hours); counts of major appliances (refrigerators, washing machines, water heaters, cooking appliances, computers, pumps and so on); solar, backup generator, cooking with grid electricity, home business, renting; socio-economic class (SEC A–E) and service area.

Timing rule. Every feature must be known at the moment of prediction, before a bill. All features are Wave 1 survey answers. Three survey fields were **excluded** because they contain or reveal the outcome: total monthly household expenditure (includes the electricity bill), units generated by solar, and the Wave 3 billing table.

1.4 Preprocessing and ethics

- Comma-formatted readings (e.g. "1,476") converted to numbers; missing readings ignored in the median.

- Missing values: floor area (26), building age (317, "don't know"); EBM bins missing natively; other models use median imputation inside the pipeline (fitted on training data only).
- Rare categories grouped (service areas outside the top 12 → "Other"; house types into five groups); one-hot encoding for the comparison models.
- **Privacy:** the data are de-identified (household IDs only). No addresses or account numbers are used, and the app holds only aggregated survey features of test households.

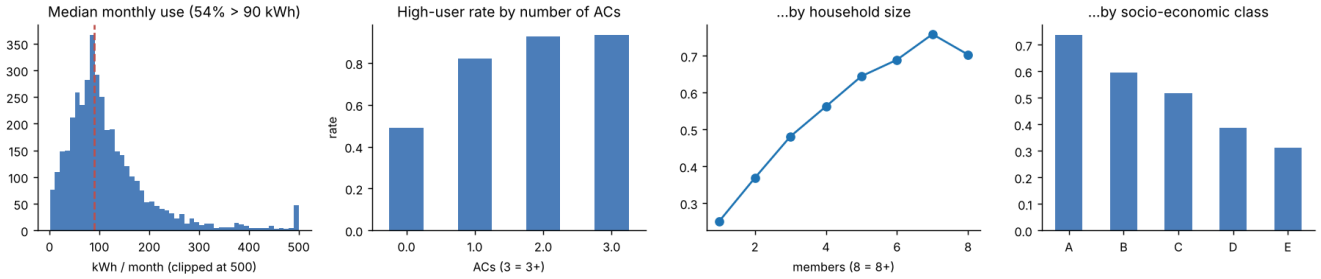


Figure 1. Target distribution and three raw relationships: high-user rate rises with ACs, household size and socio-economic class.

2. Selected algorithm: Explainable Boosting Machine

The EBM [2, 3] is a **generalised additive model with pairwise interactions** (GA²M), trained with gradient boosting:

$$\textit{logit}\, P(y = 1 \mid x) = \beta_0 + \Sigma_j\, f_j(x_j) + \Sigma_{(i,j)}\, f_{ij}(x_i, x_j)$$

- **Cyclic boosting:** features are binned; in each round, a very shallow tree is fitted on *one feature at a time* to the current residuals, with a small learning rate, cycling through all features for thousands of rounds. Each feature's trees sum into its own shape function f_j.
- **Interactions:** after the main effects, the FAST heuristic ranks feature pairs, and the top pairs get 2-D shape functions f_{ij}.
- **Bagging:** the whole procedure is repeated on bootstrap samples ("outer bags") and averaged, which smooths the shapes and gives confidence bands.

How it differs from other models. Gradient boosting, XGBoost and LightGBM grow trees over all features jointly, so the result is a black box that needs SHAP or LIME to explain. Logistic regression is additive but linear. The EBM keeps the additive, exactly decomposable structure of a GAM while learning non-linear shapes like a boosted model: each prediction is the exact sum of per-feature contributions, so its explanations are the model itself, not an approximation.

Why it suits this problem. Energy advice must be explainable to households and programme staff ("your four ACs used 40 hours a week add X"), the features are heterogeneous survey answers with missing values and skewed counts (handled natively by binning), and n ≈ 4,000 is a size where a constrained model should overfit less than deep ensembles.

Configuration (InterpretML 0.7.8): outer bags 8; tuned values: interactions 10, learning rate 0.0144, max bins 64, max leaves 3, min samples per leaf 20; early stopping on an internal validation split.

3. Model training and evaluation

Split: stratified 70 / 10 / 20: train 2,788, validation 398, test 797 households (each 54.3% positive). The test set was used once, at the end.

Tuning: randomised search with 5-fold stratified cross-validation on the training set, optimising ROC-AUC, for the EBM and five comparison models (logistic regression, decision tree, random forest, XGBoost, LightGBM). All preprocessing sits inside each pipeline, so no fold sees statistics from another. **Metrics:** ROC-AUC (threshold-free

ranking quality, the primary metric), accuracy, precision, recall and F1 at threshold 0.5, Brier score (calibration); 95% confidence intervals and paired EBM-vs-model differences from 500 bootstrap resamples of the test set.

Table 1. Test-set results. Overfit gap = training AUC − test AUC.

model	ROC-AUC	AUC CI low	AUC CI high	Accuracy	Precision	Recall	F1	Brier	CV AUC	Train AUC	Overfit gap
EBM	0.822	0.792	0.849	0.743	0.754	0.781	0.767	0.172	0.795	0.838	0.015
Logistic Regression	0.821	0.792	0.851	0.742	0.755	0.776	0.765	0.172	0.791	0.808	-0.013
Decision Tree	0.792	0.760	0.823	0.713	0.707	0.806	0.753	0.185	0.761	0.792	0.000
Random Forest	0.824	0.795	0.852	0.737	0.742	0.790	0.765	0.173	0.795	0.929	0.104
XGBoost	0.819	0.789	0.848	0.734	0.745	0.776	0.760	0.173	0.796	0.925	0.105
LightGBM	0.820	0.790	0.849	0.742	0.752	0.783	0.767	0.173	0.794	0.915	0.095

Table 2. Paired bootstrap difference in test AUC, EBM minus each model. "Significant" = 95% CI excludes 0.

model	ΔAUC (EBM − model)	95% CI	significant
Logistic Regression	+0.001	-0.010 to 0.011	no
Decision Tree	+0.030	0.013 to 0.048	yes
Random Forest	-0.002	-0.010 to 0.006	no
XGBoost	+0.003	-0.005 to 0.011	no
LightGBM	+0.003	-0.006 to 0.011	no

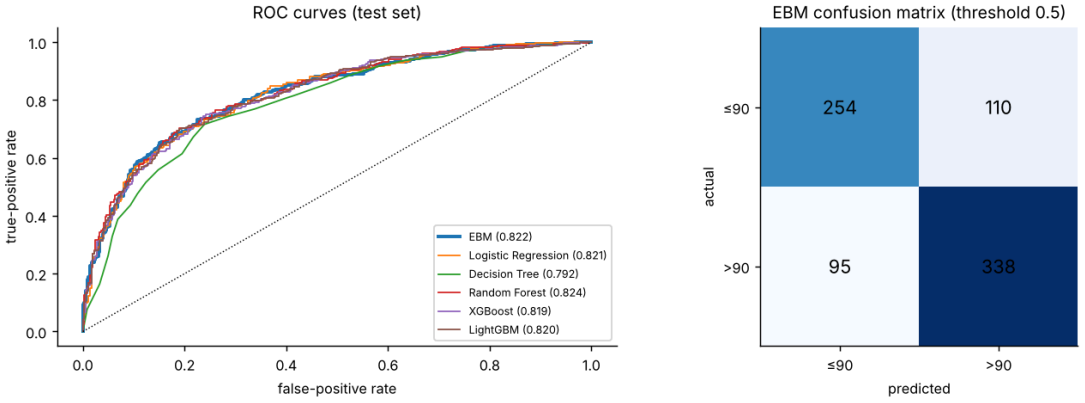


Figure 2. ROC curves of all models on the test set, and the EBM confusion matrix.

Result. The EBM (AUC 0.822) is in a statistical tie with random forest (0.824), XGBoost, LightGBM and logistic regression: all paired differences are within ±0.003 AUC and none is significant. It is significantly better than the decision tree (+0.030). The tree ensembles fit the training data far more closely (overfit gap 0.09–0.11) than the EBM (0.015). At 0.5 the EBM finds 78% of high users with 75% precision. Accuracy (about 74%) is limited by what a survey can capture: two similar homes can use very different amounts.

4. Explainability and interpretation

Three complementary methods are used: the EBM's own exact explanations (global importance, shape functions, interactions, local contributions), **SHAP** [4] (TreeSHAP on LightGBM, the strongest black box) and **LIME** [5] (local surrogate on the EBM, to test whether a model-agnostic method recovers the exact answer).

4.1 Global behaviour

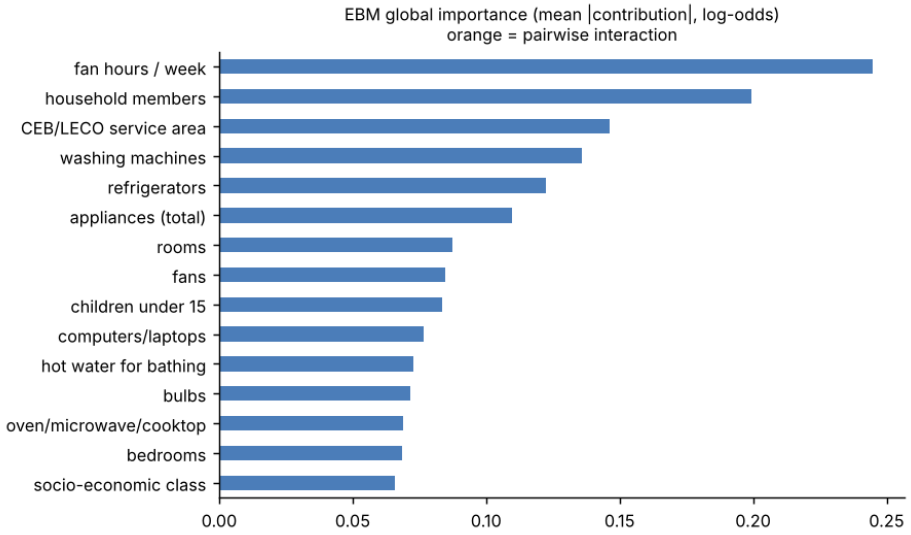


Figure 3. EBM global importance: mean absolute contribution to the log-odds across training households.

The five most influential features are **fan hours / week, household members, CEB/LECO service area, washing machines, refrigerators**. Fan hours are a proxy for both occupancy and cooling demand; household size and the main "white goods" (refrigerators, washing machines) follow. Service area matters: some areas have higher use even after appliances are accounted for (house size, local climate and income are partly bundled into area).

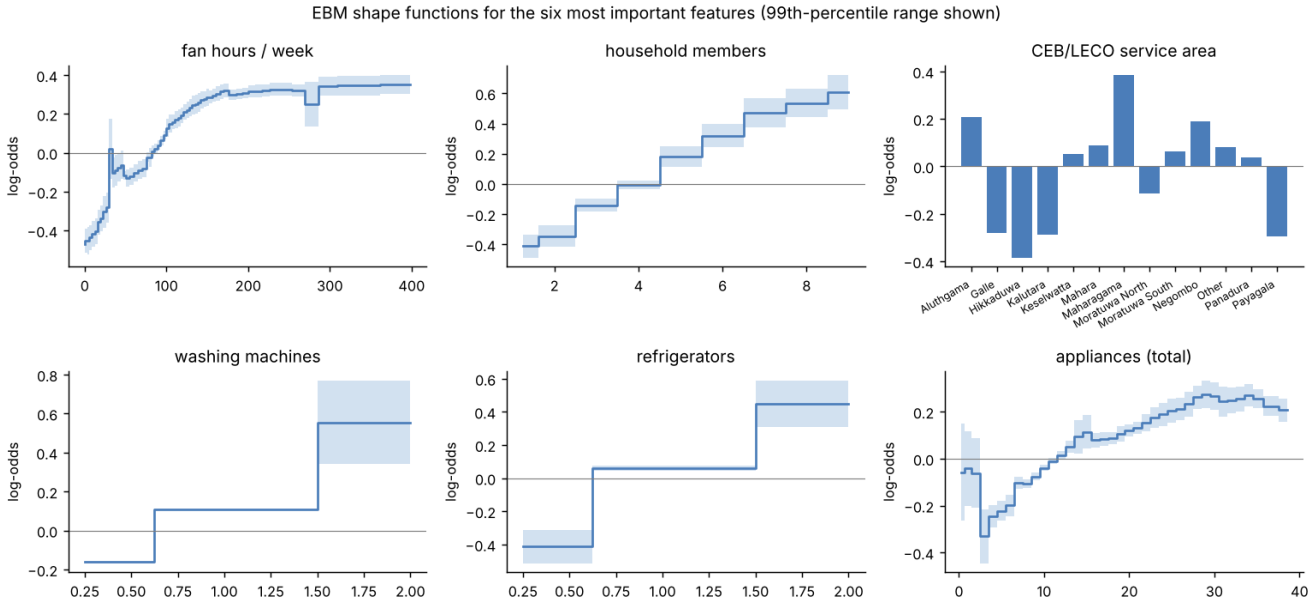


Figure 4. Shape functions: contribution (log-odds) of each value. Shaded bands = variation across the 8 outer bags.

The shapes are readable directly. Contribution rises steadily with household size (≈ −0.4 for one or two members to +0.6 for nine); fan use rises to about 150 hours a week and then plateaus; a second refrigerator or washing machine adds about +0.4. The top interaction terms are household members × refrigerators, household members × bulbs, built period (0 = pre-1980 … 5 = 2020+) × fan hours / week: for example, the effect of an extra refrigerator depends on how many people live in the home.

4.2 Local explanations

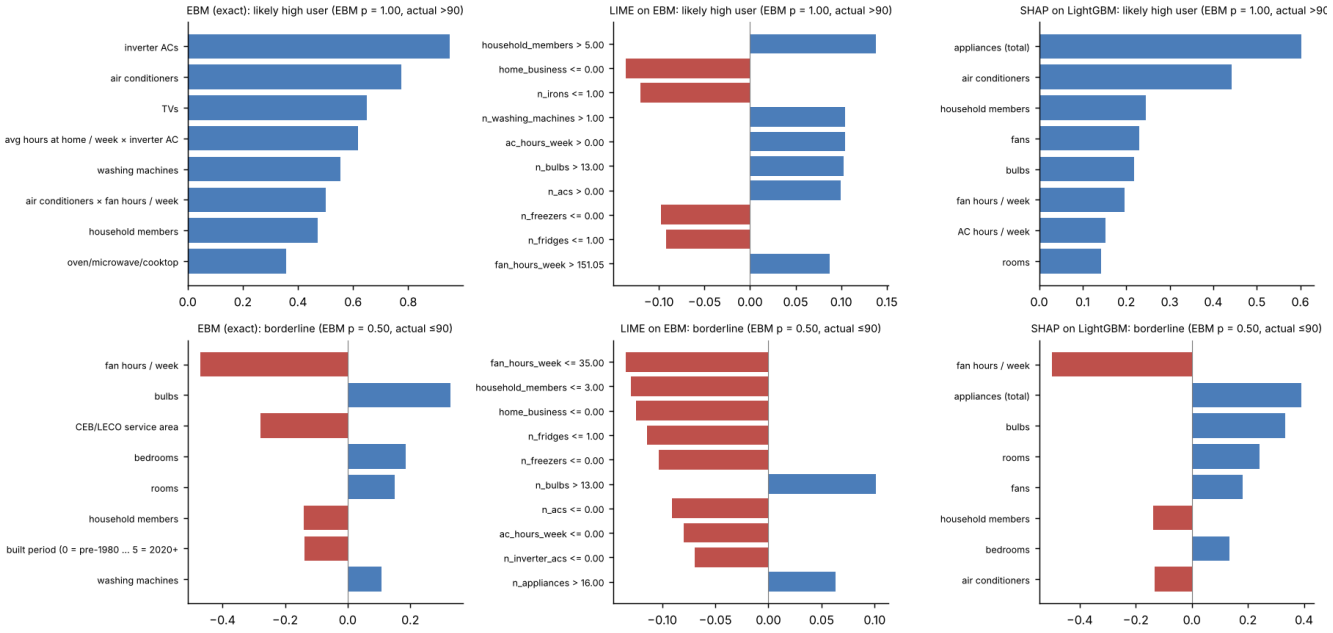


Figure 5. Two real test households explained three ways. Blue raises, red lowers the probability of being a high user.

For the clear high user (EBM p = 1.00, actual 874 kWh/month) all three methods point to air conditioning, appliances and household size. For the borderline case (p = 0.50, actual 73 kWh/month) low fan use pushes down and many bulbs push up, and the methods disagree more, as expected near the boundary.

4.3 Do the explanations agree, and are they sensible?

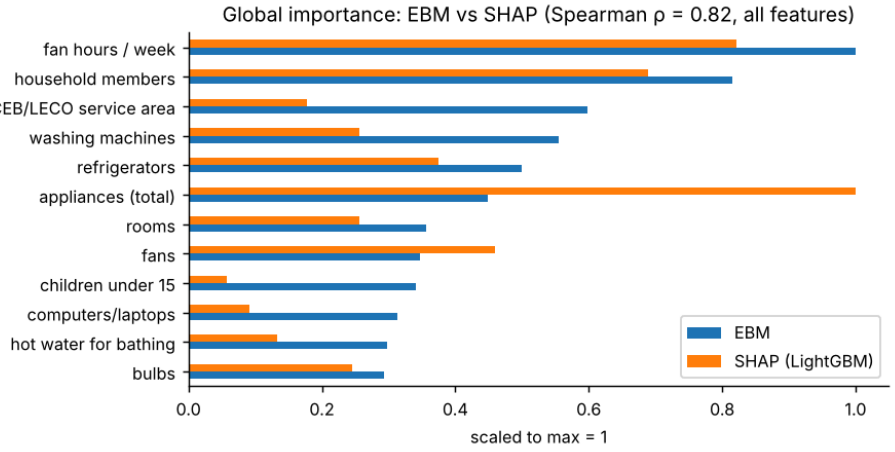


Figure 6. Global importance from the EBM and from SHAP on LightGBM, each scaled to its maximum.

- **EBM vs SHAP (global):** Spearman ρ = 0.82 over all features; 7 of the top 10 features are shared. Two different models learned the same drivers.
- **EBM vs LIME (local, 30 test households):** median ρ = 0.65. LIME recovers the EBM's main drivers but only approximately, and its top feature changed across random seeds (stable in 3/5 runs). This illustrates why an exactly interpretable model is preferable to post-hoc approximations when explanations will be acted on.
- **Domain checks (5/6 pass):**
  - ✔ More AC hours raise the odds (0 vs 40 h/week)
  - ✔ Larger households use more (2 vs 6 members)
  - ✔ Grid-powered water heating raises use (vs none)
  - ✘ On-grid solar lowers metered use (expected; NOT observed)
  - ✔ Higher socio-economic class uses more (A vs E)

- ✔ Bigger floor area uses more (600 vs 2000 sq ft)

The failed check is informative: solar households are *more* likely to be high users (about 88% of on-grid solar homes vs 51% without solar). Solar is concentrated in SEC A homes, which use more electricity, and the provider record may report import from the grid rather than net use. The model learned a genuine pattern in the data, but it is not a causal effect of installing solar, and the report flags it rather than hiding it.

5. Critical discussion

Strengths. Real, recent Sri Lankan data with an outcome from provider records rather than self-report; a strict timing rule; and a model that is as accurate as the best black box, overfits less, and gives exact explanations that domain checks and SHAP confirm.

Limitations.

- **Predictive ceiling:** AUC ≈ 0.82 means survey answers explain a lot but not everything. Behaviour (thermostat settings, actual usage of each appliance) and appliance efficiency are only partly captured.
- **Self-reported hours:** AC, fan and light hours refer to "last week" and may not represent the 12-month target window.
- **Representativeness:** mainly urban and semi-urban Western/Southern Province households from a research sample, so it may not transfer to rural or other provinces.
- **Target definition:** a 12-month median around the survey date; tariff changes in 2023–24 may have shifted behaviour.
- **Solar and metering:** as discussed in §4.3, metered consumption is not the same as total consumption for solar homes.

Fairness and ethics. Socio-economic class strongly predicts use, and errors are unequal across classes (Table 3). Lower-SEC high users are missed more often (recall 0.36 for SEC E vs 0.93 for SEC A), because high use is rarer in those groups. If the model were used to target *support*, this would under-serve poorer households; if it were used to target *charges*, it would be discriminatory. The model is therefore suitable only for screening and advice, with human review, and should never set prices or restrict supply. The EBM makes this visible: the SEC term can be inspected or removed, which a black box would hide.

Table 3. EBM error rates by socio-economic class (test set, threshold 0.5).

SEC	n	actual high-user rate	predicted rate	recall	precision	AUC
A	166	0.777	0.831	0.930	0.870	0.855
B	175	0.526	0.686	0.870	0.667	0.808
C	268	0.541	0.534	0.710	0.720	0.767
D	140	0.379	0.293	0.566	0.732	0.787
E	48	0.292	0.125	0.357	0.833	0.834

Real-world impact. A utility or a solar/efficiency programme could screen applicants with a short form and explain to each household which appliances drive its predicted use, a concrete, trustworthy message in a period of high tariffs.

6. Front end

A Streamlit app (`app.py`) is deployed on **Azure Container Apps** at https://household-electricity.redwave-d9ae0d05.eastasia.azurecontainerapps.io. A Docker image is built by GitHub Actions and published to the GitHub Container Registry. Users choose any of the six models, start from a real test household or a blank form, edit the answers and get the prediction with an explanation (EBM exact, SHAP or LIME). Other tabs show the model comparison, the EBM's global behaviour, and the ethics and fairness notes.

References

[1] ReCon SL: Residential Electricity Consumption Dataset, Sri Lanka (2025). IEEE DataPort. https://ieee-dataport.org/open-access/residential-electricity-consumption-dataset-sri-lanka-recon-sl

[2] Lou, Y., Caruana, R., Gehrke, J. & Hooker, G. (2013). Accurate intelligible models with pairwise interactions. *Proc. KDD*, 623–631.

[3] Nori, H., Jenkins, S., Koch, P. & Caruana, R. (2019). InterpretML: A unified framework for machine learning interpretability. arXiv:1909.09223.

[4] Lundberg, S. & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS*.

[5] Ribeiro, M. T., Singh, S. & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. *Proc. KDD*.

[6] Public Utilities Commission of Sri Lanka. Domestic tariff (block tariff, customer category D-1). https://www.pucsl.gov.lk/?p=8390

Tools and AI use. Python 3.12, InterpretML 0.7.8, scikit-learn 1.9, XGBoost 3.4, LightGBM 4.7, SHAP 0.54, LIME 0.2, Streamlit. Random seed 42. An AI coding assistant helped write and debug code. Modelling decisions and interpretation were reviewed by the author.